Adam Sikora

1. Algebraic Preliminaries

1.1. **Double cross product, quantum double.** In this subsection, we follow [Maj94], [KS97, Section 8.2].

A *skew pairing* of two bialgebras A, B over a field $\mathbb{K}$, is a bilinear form $\sigma : A\otimes_{\mathbb{K}} B \to \mathbb{K}$ satisfying

(1) ?e.pairing1? $$\sigma(a\otimes 1) = \epsilon_A(a), \quad \sigma(1\otimes b) = \epsilon_B(b)$$

(2) {?} $$\sigma(a\cdot a', b) = \sum \sigma(a, b_{(1)})\cdot\sigma(a', b_{(2)})$$

(3) ?e.pairing3? $$\sigma(a, b\cdot b') = \sum \sigma(a_{(2)}, b)\cdot\sigma(a_{(1)}, b').$$

(Note that a skew pairing differs from a Hopf pairing by switching $a_{(1)}$ and $a_{(2)}$ in the last identity.)

Assume now that A, B are Hopf algebras with invertible antipodes. Then any pairing σ between them has an inverse

$$\bar\sigma(a\otimes b) = \sigma(S(a), b) = \sigma(a, S^{-1}(b))$$

and the *double cross product* $B\bowtie_\sigma A$, is the algebra $B\otimes A$, with the product such that the linear embeddings $A \hookrightarrow B\otimes A, a \to 1\otimes a$ and $B \hookrightarrow B\otimes A, b \to b\otimes 1$ are algebra maps and

(4) ?e.ba? $$ba = b\otimes a$$

(5) ?e.ab? $$ab = \sum \bar\sigma(a_{(1)}\otimes b_{(1)})\sigma(a_{(3)}\otimes b_{(3)})\, b_{(2)}a_{(2)},$$

for $a \in A \subset B\otimes A$ and $b \in B \subset B\otimes A$.

Furthermore, $B\bowtie_\sigma A$ is a Hopf algebra with the coproduct

$$\nabla(b\otimes a) = \sum b_{(1)}\otimes a_{(1)}\otimes b_{(2)}\otimes a_{(2)},$$

and the antipode

$$S(b\otimes a) = S(a)S(b).$$

?⟨ss.Uq⟩? 1.2. $U_q(\mathfrak{sl}(n))$ **and its simply-connected and De Concini-Procesi versions.** Let $\check{\mathcal{U}} = \check{U}_q(\mathfrak{sl}(n))$ be the simply-connected quantized enveloping algebra of $\mathfrak{sl}(n)$, over $\mathbb{K} = \mathbb{Q}(v)$, for $n \geq 2$ where $q = v^{2n}$, following [LS24]. (For the purpose of this section, one can work over $\mathbb{Q}(v^2)$ instead.) This algebra is denoted by $U_q^P(\mathfrak{sl}(n))$ in [Gav06]. (The index n is fixed throughout this note and we suppress it for simplicity.) It is generated by $G_i^{\pm 1}, i = 1, \dots, n$, and E_i, F_i, $i = 1, \dots n-1$ with relations,

$$G_i^{\pm 1}G_j^{\pm 1} = G_j^{\pm 1}G_i^{\pm 1}, \qquad G_1G_2\dots G_n = 1,$$

(6) ?e.GEF? $$G_iE_jG_i^{-1} = q^{\delta_{i,j}-\delta_{i,j+1}}E_j, \quad G_iF_jG_i^{-1} = q^{\delta_{i,j+1}-\delta_{i,j}}F_j,$$

(7) ?e.EF? $$E_iF_j - F_jE_i = \delta_{i,j}\frac{G_iG_{i+1}^{-1} - G_i^{-1}G_{i+1}}{q-q^{-1}},$$

$$E_iE_j = E_jE_i, \quad F_iF_j = F_jF_i, \quad \text{for } |i-j| > 1$$

$$E_i^2E_j - [2]E_iE_jE_i + E_jE_i^2 = 0 = F_i^2F_j - [2]F_iF_jF_i + F_jF_i^2, \quad \text{for } |i-j| = 1.$$

for all i, j, [Gav06]. Here we use the standard notation for quantum integers:

$$[m] := \frac{q^m - q^{-m}}{q-q^{-1}} \in \mathbb{Z}[q^{\pm 1}].$$

The generators $G_1, ..., G_n$ correspond to the weight of the n-dimensional defining representation of $\mathfrak{sl}(n)$ and are denoted by $\hat K_1, ..., \hat K_n$ in [KS97]. The standard Drinfeld-Jimbo quantum groups

$\mathcal{U} = U_q(\mathfrak{sl}(n))$ are subalgebras of $\check{\mathcal{U}}$ generated by $K_i = G_i G_{i+1}^{-1}$, for $i = 1, ..., n-1$, associated with the simple roots, together with E_i's and F_i's.

In particular, we have

(8) ?e.KEF? $$K_j E_i = q^{a_{ij}} E_i K_j,$$

for all i, j, where (a_{ij}) is the A_{n-1} Cartan matrix, $a_{ii} = 2$, $a_{i,i\pm1} = -1$ and $a_{i,j} = 0$ for $|i-j| > 1$.

Following [Kas95, KS97], the coproduct and the antipode in $U_q(\mathfrak{sl}(n))$ satisfy

(9) ?e.copr? $$\Delta(K_i) = K_i \otimes K_i, \quad \Delta(E_i) = E_i \otimes K_i + 1 \otimes E_i, \quad \Delta(F_i) = F_i \otimes 1 + K_i^{-1} \otimes F_i.$$

(10) ?e.antip? $$S(K_i) = K_i^{-1}, \quad S(E_i) = -E_i K_i^{-1}, \quad S(F_i) = -K_i F_i.$$

?⟨r.Gav⟩? **Remark 1.1.** Gavarini's quantum groups have the opposite coproduct to those of [Kas95, KS97] and those used in the theory of skein algebras, eg. [LS24]. Consequently, Gavarini's formulas in this paper are transformed through the Cartan involution, which is an algebra homomorphism and coalgebra anti-homomorphism.

(11) ?e.Cartan? $$\phi(E_i) = F_i, \quad \phi(F_i) = E_i, \quad \phi(G_i) = G_i^{-1}.$$

The positive Borel subalgebra $\check{\mathcal{U}}^{\geq} \subset \check{\mathcal{U}}$ is the $\mathbb{Q}(v)$-subalgebra generated by $G_i^{\pm 1}$, for $i = 1, \dots, n$ and by E_i, for $1 \leq i < n$. Similarly, $\check{\mathcal{U}}^{\leq} \subset \check{\mathcal{U}}$ is the $\mathbb{Q}(v)$-subalgebra generated by $G_i^{\pm 1}$, for $i = 1, \dots, n$ and by F_i, for $1 \leq i < n$.

1.3. **The quantum matrix algebra.** The quantum $n \times n$ matrix algebra, $M_q(n)$, is the $\mathbb{Q}(v)$-algebra generated by variables u_{ij} subject to relations

(12) ?e.matrix1? $$u_{ik}u_{jk} = q u_{jk}u_{ik}, \quad u_{ki}u_{kj} = q u_{kj}u_{ki}, \text{ for } i < j$$

(13) ?e.matrix2? $$u_{ij}u_{kl} = u_{kl}u_{ij}, \text{for } i < k \text{ and } l < j,$$

(14) ?e.matrix3? $$u_{ij}u_{kl} - u_{kl}u_{ij} = (q - q^{-1}) u_{il}u_{kj} \text{ for } i < k \text{ and } j < l.$$

The *quantized coordinate algebra* $\mathcal{O}$ is the quotient of $M_q(n)$ (for some fixed n) by $\det_q \mathbf{u} - 1$, where $\mathbf{u} = (u_{ij})_{i,j=1}^n$; see eg. [KS97].

Let $\mathcal{O}^{\geq}$ be $\mathcal{O}$ quotiented by the 2-sided ideal generated by u_{ij} for $i > j$. Denote the images of u_{ij} in $\mathcal{O}^{\geq}$ by b_{ij}. Similarly, let

$$\mathcal{O}^{\leq} = \mathcal{O}/(u_{ij} = 0, i < j)$$

and let $c_{ij} \in \mathcal{O}^{\leq}$ be the image of u_{ij}. Then (b_{ij}) and (c_{ij}) are upper and lower triangular matrices, respectively, and the quantum determinant relation implies

(15) ?e.proddiag? $$b_{11} \cdot ... \cdot b_{nn} = 1 = c_{11} \cdot ... \cdot c_{nn}.$$

In particular, diagonal elements b_{ii}, c_{ii} are invertible.

1.4. **Gavarini's isomorphisms.** Let $r = q - q^{-1}$.

?⟨p.Gavarini⟩? **Proposition 1.2** ([Gav06] Sec. 3.4.). *There are isomorphisms of Hopf $\mathbb{Q}(v)$-algebras*

(16) ?e.zeta_leq? $\zeta_{\geq} : \check{\mathcal{U}}^{\geq} \to \mathcal{O}^{\leq}$, *given by* $G_i \to c_{ii}$, $E_i \to r^{-1} c_{i+1,i+1}^{-1} c_{i+1,i}$

(17) ?e.zeta_geq? $\zeta_{\leq} : \check{\mathcal{U}}^{\leq} \to \mathcal{O}^{\geq}$, *given by* $G_i \to b_{ii}^{-1}$, $F_i \to -r^{-1} b_{i,i+1} b_{i+1,i+1}^{-1}$

whose inverses are

$$\zeta_{\geq}^{-1} : \mathcal{O}^{\leq} \xrightarrow{\cong} \check{\mathcal{U}}^{\geq}, \text{ given by } c_{ii} \to G_i,\ c_{i+1,i} \to r G_{i+1} E_i$$

$$\zeta_{\leq}^{-1} : \mathcal{O}^{\geq} \xrightarrow{\cong} \check{\mathcal{U}}^{\leq}, \text{ given by } b_{ii} \to G_i^{-1},\ b_{i,i+1} \to -r F_i G_{i+1}^{-1}.$$

(Following Remark 1.1, we make Gavarini's ζ consistent with our conventions by precomposing his ζ with the Cartan involution.)

?⟨s.Rform⟩? 1.5. **The R-form and the Drinfeld-Tanisaki skew-pairing.** Recall that $\mathcal{O}$ has an R-form $\rho : \mathcal{O} \otimes \mathcal{O} \to \mathbb{Q}(v)$ defined by

(18) ?e.coR? $$\rho(x, y) = \langle x \otimes y, R \rangle,$$

where $\langle x \otimes y, u \otimes w \rangle$ is the product of the duality pairings $\langle x, u \rangle \cdot \langle y, w \rangle$. Equivalently,

$$\rho(u_{ij} \otimes u_{kl}) = R_{jl}^{ik}.$$

Proposition 1.3. *The bilinear form $\rho : \mathcal{O} \otimes \mathcal{O} \to \mathbb{Q}(v)$ is a skew pairing and it descends to a skew pairing*

$$\sigma : \mathcal{O}^{\leq} \otimes \mathcal{O}^{\geq} \to \mathbb{Q}(v).$$

Proof. We have

(19) ?eq.rho1? $$\rho(u_{ij} \otimes u_{kl}) = R_{jl}^{ik} = q^{-\frac{1}{n}} \left(q^{\delta_{i,k}} \delta_{i,j} \delta_{k,l} + (q - q^{-1}) \delta_{j<i} \delta_{k,j} \delta_{i,l} \right)$$

(20) ?eq.rho2? $$= q^{-\frac{1}{n}} \left(q^{\delta_{i,k}} \delta_{i,j} \delta_{k,l} + (q - q^{-1}) \delta_{k<l} \delta_{k,j} \delta_{i,l} \right)$$

where $q^{\frac{1}{n}} = v^2$ and $\delta_{j<i}$ is either 1 or 0 depending on whether $j < i$ or not, [KS97, Sec. 8.4.2(60)]. If $i < j$ then $\rho(u_{ij} \otimes u_{kl}) = 0$ by (19). Now, the skew-pairing identities immediately imply that $\rho(x \otimes u_{kl}) = 0$ for all x in the two-sided ideal generated by such u_{ij}. (Since $\Delta(u_{ij}) = \sum_m u_{im} \otimes u_{mj}$ and every m satisfies $i < m$ or $m < j$, identity (3) and induction on the degree of y imply that $\rho(u_{ij} \otimes y) = 0$ for all $y \in \mathcal{O}$ and $i < j$.) Hence, ρ factors to

$$\mathcal{O}^{\leq} \otimes \mathcal{O} \to \mathbb{Q}(v).$$

If $k > l$ then $\rho(u_{ij} \otimes u_{kl}) = 0$ by (20) and, consequently, the above factors to

$$\sigma : \mathcal{O}^{\leq} \otimes \mathcal{O}^{\geq} \to \mathbb{Q}(v).$$

□

By the isomorphism of Proposition 1.2, σ defines a skew-pairing

$$\sigma' : \check{\mathcal{U}}^{\geq} \otimes \check{\mathcal{U}}^{\leq} \to \mathbb{Q}(v).$$

We also have the Drinfeld-Tanisaki skew-pairing between the quantum groups of the upper and lower Borel groups:

$$\langle \cdot, \cdot \rangle : \check{\mathcal{U}}^{\geq} \otimes \check{\mathcal{U}}^{\leq} \to \mathbb{Q}(v),$$

(21) ?e.standardpairing? $\langle G_i, G_j \rangle = q^{\frac{1}{n} - \delta_{i,j}}$, $\langle E_i, F_j \rangle = -\delta_{i,j}/r$, $\langle G_i, F_j \rangle = \langle E_j, G_i \rangle = 0$,

for all i, j, cf. [KS97, (6.82)].

Proposition 1.4. *σ' is the Drinfeld-Tanisaki skew-pairing $\langle\cdot,\cdot\rangle : \check{\mathcal{U}}^{\geq} \otimes \check{\mathcal{U}}^{\leq} \to \mathbb{Q}(v)$.*

Proof. We need to show that

$$\sigma'(x,y) = \sigma\big(\zeta_{\geq}(x), \zeta_{\leq}(y)\big)$$

satisfies

$$\sigma'(G_i, G_j) = q^{\frac{1}{n}-\delta_{ij}}, \qquad \sigma'(E_i, F_j) = -\delta_{ij}/r$$

and

$$\sigma'(G_i, F_j) = 0 = \sigma'(E_j, G_i).$$

Recall that the R-form on the matrix generators is

$$\rho(u_{ij}, u_{kl}) = v^{-2}\left(q^{\delta_{ik}}\delta_{ij}\delta_{kl} + r\,\delta_{j<i}\delta_{k,j}\delta_{i,l}\right).$$

After passing to the lower- and upper-triangular quotients, this gives

$$\sigma(c_{aa}, b_{bb}) = v^{-2}\ q^{\delta_{ab}}, \quad \sigma(c_{i+1,i}, b_{j,j+1}) = v^{-2} r\,\delta_{ij} \quad \sigma(c_{aa}, b_{j,j+1}) = 0, \text{ and } \sigma(c_{i+1,i}, b_{aa}) = 0.$$

The diagonal elements c_{aa} and b_{bb} are group-like. Consequently,

$$\sigma(c_{aa}, b_{bb}^{-1}) = \sigma(c_{aa}, b_{bb})^{-1} = v^2 q^{-\delta_{ab}} = q^{\frac{1}{n}-\delta_{ab}}.$$

Since

$$\zeta_{\geq}(G_a) = c_{aa}, \qquad \zeta_{\leq}(G_b) = b_{bb}^{-1},$$

we obtain

$$\sigma'(G_a, G_b) = q^{\frac{1}{n}-\delta_{ab}}.$$

The same formulas immediately imply

$$\sigma'(G_a, F_j) = 0 \qquad \text{and} \qquad \sigma'(E_i, G_b) = 0.$$

Indeed, the images of F_j and E_i are scalar multiples of an off-diagonal matrix generator times a diagonal element, and the pairing of a diagonal generator with an off-diagonal generator vanishes.

It remains to calculate $\sigma'(E_i, F_j)$. Set

$$C_i = c_{i+1,i+1}, \qquad X_i = c_{i+1,i},$$

and

$$B_j = b_{j,j+1}, \qquad D_j = b_{j+1,j+1}.$$

Gavarini's isomorphisms give

$$\zeta_{\geq}(E_i) = \frac{C_i^{-1} X_i}{r}, \qquad \zeta_{\leq}(F_j) = -\frac{B_j D_j^{-1}}{r}.$$

The relevant coproducts in the triangular coordinate algebras are

$$\Delta(X_i) = X_i \otimes c_{ii} + C_i \otimes X_i$$

and

$$\Delta(B_j) = b_{jj} \otimes B_j + B_j \otimes D_j.$$

Since C_i and D_j are group-like, it follows that

$$\Delta(B_j D_j^{-1}) = b_{jj} D_j^{-1} \otimes B_j D_j^{-1} + B_j D_j^{-1} \otimes 1.$$

Using the first skew-pairing identity, we obtain

$$\sigma(C_i^{-1} X_i, B_j D_j^{-1}) = \sigma(C_i^{-1}, b_{jj} D_j^{-1}) \sigma(X_i, B_j D_j^{-1}),$$

because the other coproduct term contains the factor $\epsilon(X_i) = 0$.

We first calculate the second factor. The second skew-pairing identity and the coproduct of X_i give

$$\sigma(X_i, B_j D_j^{-1}) = \sigma(c_{ii}, B_j)\sigma(X_i, D_j^{-1}) + \sigma(X_i, B_j)\sigma(C_i, D_j^{-1}).$$

The first summand vanishes. Furthermore,

$$\sigma(X_i, B_j) = v^{-2} r\, \delta_{ij}.$$

If $i = j$, then

$$\sigma(C_i, D_i) = \sigma(c_{i+1,i+1}, b_{i+1,i+1}) = v^{-2} q = q^{\frac{n-1}{n}},$$

and hence

$$\sigma(C_i, D_i^{-1}) = q^{-\frac{n-1}{n}}.$$

It follows that

$$\sigma(X_i, B_j D_j^{-1}) = \frac{r}{q}\, \delta_{ij}.$$

For the first factor, only the case $i = j$ remains relevant. In that case,

$$\sigma(C_i^{-1}, b_{ii} D_i^{-1}) = \sigma(C_i^{-1}, b_{ii})\sigma(C_i^{-1}, D_i^{-1}) = v^2 \cdot v^{-2} q = q.$$

Here we used

$$\sigma(C_i, b_{ii}) = v^{-2}$$

and, for group-like elements,

$$\sigma(C_i^{-1}, D_i^{-1}) = \sigma(C_i, D_i) = q^{\frac{n-1}{n}}.$$

Therefore,

$$\sigma(C_i^{-1} X_i, B_j D_j^{-1}) = r\, \delta_{ij}.$$

Including the normalization constants in Gavarini's maps gives

$$\sigma'(E_i, F_j) = \sigma\left(\frac{C_i^{-1} X_i}{r}, -\frac{B_j D_j^{-1}}{r}\right) = -\frac{1}{r^2}\, \sigma(C_i^{-1} X_i, B_j D_j^{-1}) = -\frac{\delta_{ij}}{r}.$$

We have therefore shown that σ' agrees on the generators with the Drinfeld–Tanisaki skew pairing. Since both forms are skew pairings and the two Borel algebras are generated by these elements, the skew-pairing identities determine their values on arbitrary products and the statement follows. □

2. Skein Algebras

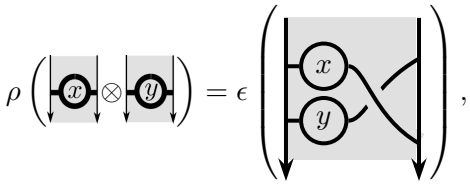
?⟨s.skeinalg⟩?

The stated skein algebras $\mathscr{S}_n(\Sigma)$ of punctured-bordered surfaces Σ were introduced in [Le18], [Hig23], [LS24] for $n = 2$, $n = 3$ and all n, respectively. We consider them here over the base field $\mathbb{K} = \mathbb{Q}(v)$, where v is as in [LS24]. Since n is fixed, we denote them by $\mathscr{S}(\Sigma)$ for simplicity.

The stated skein algebras of the bigon, $\mathscr{S}(\mathfrak{B})$, were shown to coincide with quantized coordinate algebra, $\mathcal{O} = \mathcal{O}_q(\mathrm{SL}(n))$, on $\mathrm{SL}(n)$, for $n = 2, 3$, and all n in [CL22], [Hig23], and [LS24], respectively, through an isomorphism sending arcs of Fig. 2(left) to elements u_{ij}. (Whenever identifying these two algebras we will assume this isomorphism. Bigons will be depicted having their distinguished ideal points at the bottom.)

The geometric description of the R-form ρ of Sec. 1.5 in this skein setting was given in [LS24, Theorem 6.4]:

(22) ?e.cobraid?



2.1. **Stated skein algebra of the punctured bigon.** In [LS24] we have introduced saturated systems of arcs in pb surfaces Σ and proved that they induce isomorphisms between $\mathscr{S}(\Sigma)$ and tensor powers of $\mathcal{O}$. The arcs in Fig. 1 form a saturated system in the punctured bigon $\mathfrak{B}^*$. Denote by $\mathfrak{B}^+$ and by $\mathfrak{B}^-$ the tubular neighborhoods of the lower and of the upper arc, respectively. By [LS24, Thm. 8.8], their embeddings into $\mathfrak{B}^*$ define a linear isomorphism

(23) ?e.SfB*?
$$\Phi : \mathcal{O} \otimes \mathcal{O} = \mathscr{S}(\mathfrak{B}^+) \otimes \mathscr{S}(\mathfrak{B}^-) \to \mathscr{S}(\mathfrak{B}^*),$$

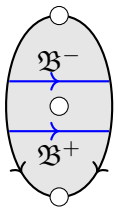


FIGURE 1. Punctured bigon, $\mathfrak{B}^*$, with a saturated system of arcs.

Identifying $\mathscr{S}(\mathfrak{B}^+)$ and $\mathscr{S}(\mathfrak{B}^-)$ with $\mathcal{O}$, the skew pairing ρ becomes,

$$\rho : \mathscr{S}(\mathfrak{B}^-) \otimes \mathscr{S}(\mathfrak{B}^+) \to \mathbb{Q}(v),$$

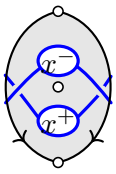
making the above a skein interpretation of the quantum double. Specifically:

?⟨p.QuantD⟩? **Proposition 2.1.** (23) *is an algebra isomorphism* $\mathscr{S}(\mathfrak{B}^+) \bowtie_\rho \mathscr{S}(\mathfrak{B}^-) \to \mathscr{S}(\mathfrak{B}^*)$.

Proof. The statement follows from

(24) ?e.ba2?
$$\Phi(x^+ \cdot x^-) = \Phi(x^+) \cdot \Phi(x^-),$$

(25) ?e.ab2?
$$\Phi(x^- \cdot x^+) = \quad = \sum x^+_{(2)} x^-_{(2)} \cdot \overline{\rho}(x^-_{(1)} \otimes x^+_{(1)}) \rho(x^-_{(3)} \otimes x^+_{(3)}),$$



for all $x^- \in \mathscr{S}(\mathfrak{B}^-)$ and $x^+ \in \mathscr{S}(\mathfrak{B}^+)$ by the geometric interpretation of ρ, Eq. (22). This product agrees with that of the quantum double. □

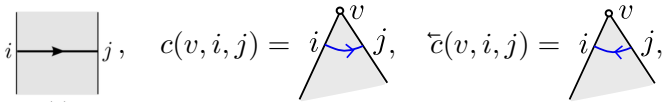


FIGURE 2. Left: A stated arc in $\mathscr{S}(\mathfrak{B})$ corresponding to $u_{ij} \in \mathcal{O}$. Center and Right: Stated v-corner arcs.

2.2. **Reduced stated skein algebras.** For an ideal boundary vertex v of a pb surface, a *v-corner arc* is a non-boundary-parallel simple arc that can be isotoped into an arbitrary neighborhood of v. We denote such stated arcs either by $c(v,i,j)$ or by $\tilde{c}(v,i,j)$ depending on their orientation, as in Fig. 2.

The quotient of the stated skein algebra $\mathscr{S}(\Sigma)$ by the 2-sided ideal $\mathcal{I}(\Sigma, v)$ generated by the arcs $c(v,i,j)$, for $i<j$, for a certain ideal boundary vertex v will be denoted by $\overline{\mathscr{S}}(\Sigma, v)$ and called the *reduced skein algebra at v*. The above-mentioned arcs are called *bad*.

The quotient of $\mathscr{S}(\Sigma)$ by the 2-sided ideal generated by the arcs $c(v,i,j)$, for $i<j$, and all ideal boundary vertices v, which do not lie in monogons, is denoted by $\overline{\mathscr{S}}(\Sigma)$ and called the *reduced skein algebra*, [CL22, LY22, LY23].

3. Skein Realization of the quantum double $\check{\mathcal{U}}^{\leq} \bowtie_{\sigma'} \check{\mathcal{U}}^{\geq}$

Let us denote the top and the bottom ideal boundary points of $\mathfrak{B}^*$ by $-\infty$ and $+\infty$, respectively (so that they are consistent with the notation $\mathfrak{B}^{\pm}$ above). Then the isomorphism Φ of Proposition 2.1 maps $\mathcal{I}(\mathfrak{B}^+, +\infty) \otimes \mathscr{S}(\mathfrak{B}^-) + \mathscr{S}(\mathfrak{B}^+) \otimes \mathcal{I}(\mathfrak{B}^-, -\infty)$ onto the ideal of bad arcs of $S(\mathfrak{B}^*)$ and, consequently, Φ factors to

$$\overline{\mathscr{S}}(\mathfrak{B}^+, +\infty) \bowtie_{\rho} \overline{\mathscr{S}}(\mathfrak{B}^-, -\infty) \to \overline{\mathscr{S}}(\mathfrak{B}^*).$$

Note that $\mathscr{S}(\mathfrak{B}) \simeq \mathcal{O}$ composed with $\zeta_{\leq}, \zeta_{\geq}$ of (16)-(17), descend to isomorphisms

$$\overline{\mathscr{S}}(\mathfrak{B}^+, +\infty) \simeq \mathcal{O}^{\geq} \simeq \check{\mathcal{U}}^{\leq}, \qquad \overline{\mathscr{S}}(\mathfrak{B}^-, -\infty) \simeq \mathcal{O}^{\leq} \simeq \check{\mathcal{U}}^{\geq}.$$

Corollary 3.1 (Cmp. [Kor] for $n=2$)**.** *The above isomorphisms define an isomorphism Ψ from the quantum double $\check{\mathcal{U}}^{\leq} \bowtie_{\sigma'} \check{\mathcal{U}}^{\geq}$ to $\overline{\mathscr{S}}(\mathfrak{B}^*)$, such that*

$$\Psi(G_i^{\geq}) = c_{ii}, \quad \Psi(G_i^{\leq}) = b_{ii}^{-1}, \quad \Psi(E_i) = r^{-1} c_{i+1,i+1}^{-1} c_{i+1,i}, \quad \text{and} \quad \Psi(F_i) = -r^{-1} b_{i,i+1} b_{i+1,i+1}^{-1},$$

for $1 \leq i < n$, where $G_i^{\geq}, G_i^{\leq}$ are the copies of G_i in $\check{\mathcal{U}}^{\geq}$ and in $\check{\mathcal{U}}^{\leq}$, respectively and the elements $b_{i,j}, c_{kl}$ defined earlier in $\mathcal{O}^{\geq}$ and $\mathcal{O}^{\leq}$ correspond to elements depicted below.

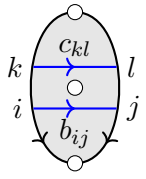


FIGURE 3. The arcs b_{ij}, c_{kl} in $\mathscr{S}(\mathfrak{B}^*)$.

4. Construction of $\check{\mathcal{U}}$

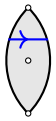
The embeddings $\check{\mathcal{U}}^{\geq}, \check{\mathcal{U}}^{\leq} \hookrightarrow \check{\mathcal{U}}$ extend to a Hopf algebra isomorphism

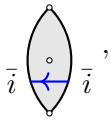
$$(\check{\mathcal{U}}^{\leq} \bowtie_{\sigma'} \check{\mathcal{U}}^{\geq})/I \to \check{\mathcal{U}},$$

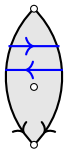
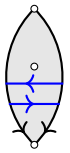
where I is generated by $G_i^{\geq} - G_i^{\leq}$ for $i = 1, ..., n$, cf. [KS97, 8.2.4 Cor. 15].

Consequently,

?⟨c.UcZ-PP21⟩? **Corollary 4.1** (Cmp. [Kor] for $n = 2$)**.** *The above isomorphism descends to an isomorphism* $\Psi : \check{\mathcal{U}} \to \overline{\mathscr{S}}(\mathfrak{B}^*)/I$, *where* I *is the ideal generated by* $b_{ii}c_{ii} - 1$ *for* $i = 1, ..., n$.

By [LS24, Eqs. (47) and (57)], it is easy to see that $b_{i,i}^{-1}$ coincides with $b_{\bar{i},\bar{i}}$ with reversed orientation. Hence, under the above isomorphisms the Serre-style generators of $\check{\mathcal{U}}$ are $G_i \simeq$  $=$


,

$$E_i = r^{-1} \cdot$$  , $$F_i = -r^{-1} \cdot$$  ,

for $j = i + 1$.

References

CL [CL22] Francesco Costantino and Thang T. Q. Lê, *Stated skein algebras of surfaces*, J. Eur. Math. Soc. (JEMS) **24** (2022), no. 12, 4063–4142.

Gavarini2 [Gav06] Fabio Gavarini, *Presentation by Borel subalgebras and Chevalley generators for quantum enveloping algebras*, Proc. Edinb. Math. Soc. (2) **49** (2006), no. 2, 291–308.

Higgins [Hig23] Vijay Higgins, *Triangular decomposition of sl_3 skein algebras*, Quantum Topol. **14** (2023), no. 1, 1–63, arXiv:2008.09419.

?HL? [HL16] Kazuo Habiro and Thang T. Q. Lê, *Unified quantum invariants for integral homology spheres associated with simple Lie algebras*, Geom. Topol. **20** (2016), no. 5, 2687–2835.

?Jantzen? [Jan96] Jens Carsten Jantzen, *Lectures on quantum groups*, Graduate Studies in Mathematics, vol. 6, American Mathematical Society, Providence, RI, 1996.

Kass [Kas95] Christian Kassel, *Quantum groups*, Graduate Texts in Mathematics, vol. 155, Springer-Verlag, New York, 1995.

Kor [Kor] Julien Korinman, *Quantum Holonomic Link Invariants Derived From Stated Skein Algebras*, arXiv: 1907.01732.

KS [KS97] Anatoli Klimyk and Konrad Schmüdgen, *Quantum groups and their representations*, Texts and Monographs in Physics, Springer-Verlag, Berlin, 1997.

e:triangulation [Le18] Thang T. Q. Le, *Triangular decomposition of skein algebras*, Quantum Topol. **9** (2018), no. 3, 591–632.

LS [LS24] Thang T. Q. Lê and Adam S. Sikora, *Stated* SL(n)*-skein modules and algebras*, J. Topol. **17** (2024), no. 3, Paper No. e12350, 93.

?LS2? [LS26] ———, *Skein approach to quantum doubles, L-operators, and quantum Killing forms*, In preparation.

LY2 [LY22] Thang T. Q. Le and Tao Yu, *Quantum traces and embeddings of stated skein algebras into quantum tori*, Selecta Math. (N.S.) **28** (2022), no. 4, Paper No. 66, 48.

LY3 [LY23] ———, *Quantum traces for sl_n-skein algebras*, 2023.

Majid2 [Maj94] Shahn Majid, *Some remarks on the quantum double*, vol. 44, 1994, Quantum groups and physics (Prague, 1994), pp. 1059–1071.